**AD 502 – Machine Learning**

**Pre-Reading Material – Lecture 32**

**Introduction to Ensemble Learning-II**

## Types of Ensemble Learning

Ensemble learning includes several techniques for combining multiple machine learning models. The most common approaches are voting, averaging, bagging, boosting, and stacking.

Each approach uses a different strategy to combine predictions.

## Voting

Voting is primarily used for classification problems. In this approach, multiple classifiers predict a class label, and their predictions are combined to determine the final class.

For example, three classifiers predict whether a transaction is fraudulent:

- Decision Tree: Fraud
- Logistic Regression: Fraud
- Support Vector Machine: Not Fraud

The majority prediction is Fraud. Therefore, the ensemble classifies the transaction as fraudulent.

Voting can be performed using hard voting or soft voting.

## Averaging

Averaging is commonly used in regression problems, where the objective is to predict a numerical value.

In this technique, the numerical predictions of multiple models are combined, usually by calculating their arithmetic mean.


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For example, three models predict electricity demand as 100 MW, 110 MW, and 105 MW. Their average prediction is 105 MW.

Averaging may reduce prediction variability when the models make different errors.

## Bagging

Bagging stands for Bootstrap Aggregating. It involves training multiple models on different bootstrap samples of the training dataset and combining their predictions.

A bootstrap sample is created by randomly selecting observations from the original dataset with replacement.

The main objective of bagging is to improve stability and reduce variance. Random Forest is a well-known example of a bagging-based ensemble method.

## Boosting

Boosting is an ensemble technique in which models are trained sequentially. Each new model attempts to improve the performance of the existing ensemble by focusing on errors or difficult observations.

Examples include AdaBoost, Gradient Boosting, and XGBoost.

Boosting can produce strong predictive performance, but it may require careful tuning to avoid overfitting.

## Stacking

Stacking, or stacked generalization, combines the predictions of multiple base models using another model called a meta-learner.

For example, the predictions of a Decision Tree, Random Forest, and Support Vector Machine can be provided as inputs to a Logistic Regression meta-learner, which generates the final classification.

Stacking can learn how to combine the base models rather than relying only on a fixed voting or averaging rule.


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## Comparison of Ensemble Techniques

| Technique | Basic principle | Common application |
|---|---|---|
| Voting | Combines predicted class labels or class probabilities | Classification |
| Averaging | Combines numerical predictions | Regression |
| Bagging | Trains models on bootstrap samples | Classification and regression |
| Boosting | Trains models sequentially to improve predictions | Classification and regression |
| Stacking | Uses a meta-model to combine predictions | Classification and regression |

## Important Considerations

The success of an ensemble depends on several factors:

- Accuracy of base models: Individual models should provide useful predictions.
- Diversity: Models that make different errors can complement one another.
- Computational cost: Training multiple models may require additional resources.
- Selection of combination technique: Voting, averaging, and other methods are suitable for different prediction tasks.

## Summary

Ensemble learning provides several strategies for combining machine learning models. Voting is commonly used for classification, averaging for numerical predictions, bagging for reducing variance, boosting for sequential error correction, and stacking for learning an effective combination of base-model predictions.


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